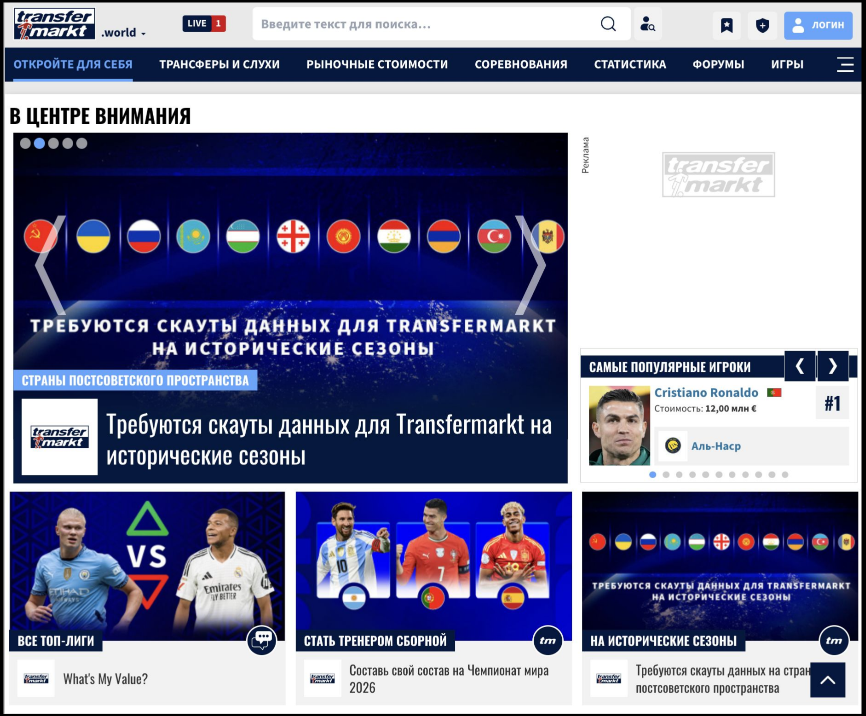
Frontier AI

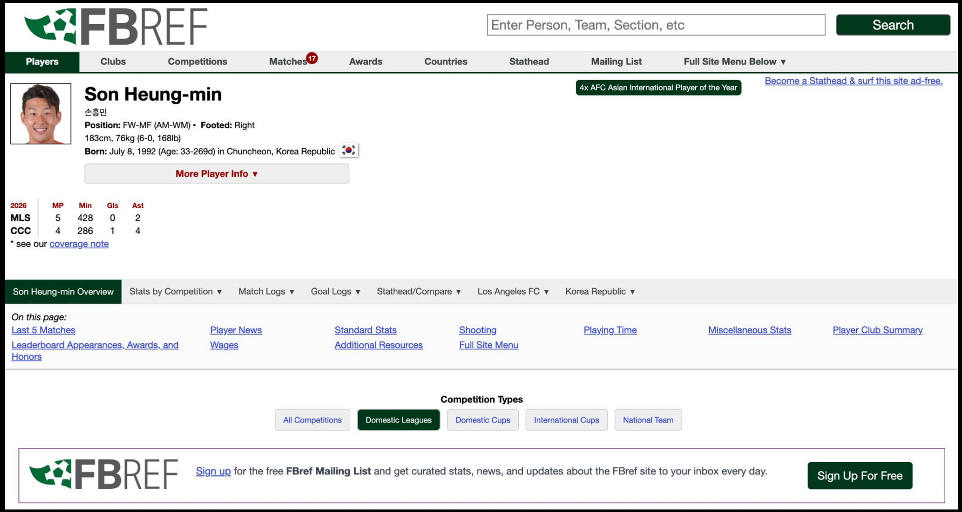
Макарчук Алексей

Лекция 8. Получение общих эмбеддингов в футболе

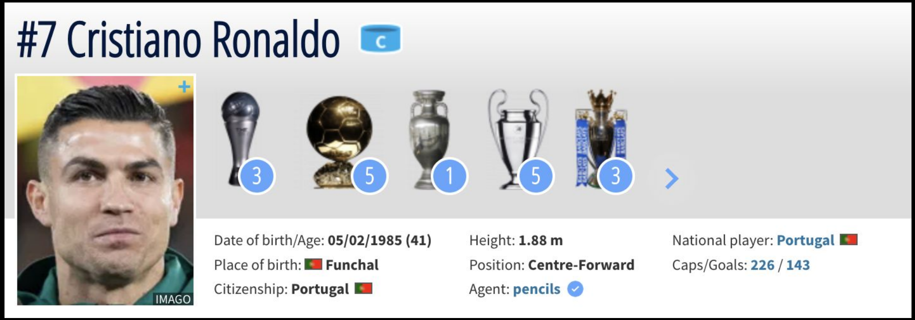
SBER TECH

Профильные источники





Профильные источники



STATS 25/26

Compact | Detailed

Competition												
Saudi Pro League	23	23	2	-	-	7	1	-	-	6	87'	2003'
Saudi Super Cup	2	1	1	-	-	1	-	-	-	1	180'	180'
AFC Champions League Two	1	-	1	-	-	1	-	-	-	-	-	45'
King's Cup	1	-	-	-	-	-	-	-	-	-	-	90'
Total 25/26:	27	24	4	-	-	9	1	-	-	7	97'	2,318'

RisingBALLER (october 2024)

StatsBomb Conference 2024, Research Stage.

RisingBALLER: A Player is a Token, a Match is a Sentence–A Path Towards a Foundational Model for Football Players Data Analytics

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10.00943v1 [cs.LG] 1 Oct 2024

Abstract

In this paper, I introduce RisingBALLER, the first publicly available approach that leverages a transformer model trained on football match data to learn match-specific player representations. Drawing inspiration from advances in language modeling, RisingBALLER treats each football match as a unique sequence in which players serve as tokens, with their embeddings shaped by the specific context of the match. Through the use of masked player prediction (MPP) as a pre-training task, RisingBALLER learns foundational features for football player representations, similar to how language models learn semantic features for text representations. As a downstream task, I introduce next match statistics prediction (NMSP) to showcase the effectiveness of the learned player embeddings. The NMSP model surpasses a strong baseline commonly used for performance forecasting within the community. Furthermore, I conduct an in-depth analysis to demonstrate how RisingBALLER's learned embeddings can be used in various football analytics tasks, such as producing meaningful positional features that capture the essence and variety of player roles beyond rigid x,y coordinates, team cohesion estimation, and similar player retrieval for more effective data-driven scouting. More than a simple machine learning model, RisingBALLER is a comprehensive framework designed to transform football data analytics by learning high-level foundational features for players, taking into account the context of each match. It offers a deeper understanding of football players beyond individual statistics.

Akedjou Achraff Adjileye - RisingBALLER:A Player is a Token, a Match is a Sentence–A Path Towards a Foundational Model for Football Players Data Analytics

Lorenzo Cascioli et al - Boot It: A Pragmatic Alternative To Build Up Play

Michael Gleeson - Modelling Team Play Style Using Tracking Data to Evaluate the Effectiveness of Variance in Tactical Behaviour

Mark Carter - Footedness: How important is it?

Edoardo Ghezzi and Hadi Sotudeh - Match Phases in Practice

Pablo Galaz-Cares - Optimizing Football Squad Planning: Balancing Competitive Success and Financial Sustainability

Daniel Pritchard - Risks and rewards of executing transition events from settled possession.

Oleg Durygin, Daniil Gumin, Egor Gumin - The coach's dilemma: how a yellow card can change the course of a match.

Berta Plandolit López - Balancing pressing intensity and defensive solidity against different passing styles

This is an archive of the article which was originally published at:
https://statsbomb.com/news/statsbomb-conference-2024-research-papers/

Dataset

Task

Build **contextual representations** of football players from match data, by analogy with language models:

- Match ≈ sentence
- Player ≈ token

Pretraining:
Masked Player Prediction (MPP) — we mask 25% of players in a match and predict who is hidden.

Data

Parameter	Value
Source	StatsBomb open data
Season	2015–2016
Matches	≈ 1 800
Unique players	≈ 2 600
Stats per player	39 features

Examples of statistics:

Passes · Shots · Interceptions
Dribbling · Fouls · Goalkeeping
Blocks, clearances, tackles, counter-pressing

Data source: github.com/statsbomb/open-data

Dataset

I collected 39 statistics from the StatsBomb data to represent each player. These include:

- 10 statistics related to passing: pass total, pass cross, pass cut back, pass shot assist, pass goal assist, pass no touch, pass interception, pass incomplete, pass offside, pass through ball.
- 9 statistics related to shots: shot total, shot statsbomb xg, shot corner, shot free kick, shot open play, shot penalty, shot saved, shot off target, shot blocked, shot goal.
- 3 statistics related to interceptions: interception total, interception won, interception lost.
- 3 statistics related to dribbles: dribble total, dribble complete, dribble incomplete.
- 5 statistics related to fouls: foul won total, foul won penalty, foul committed total, foul committed penalty, foul committed yellow card, foul committed red card.
- 3 statistics related to goalkeeping: goalkeeper goal conceded, goalkeeper save, goalkeeper shot faced.
- additionally: block total, clearance total, ball recovery total and counterpress total.

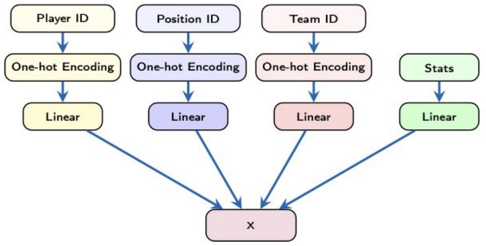
Dataset

In total, I generated data for 1,792 matches, representing 2,600 unique players (the player vocabulary) from 98 unique teams. For the masked player prediction (MPP) task, I added an ID to represent the masking token and an ID for the padding token, allowing all inputs to have the same dimensions (a fixed number of players per match) for batch computation. To compute team-level stats for the next match statistics prediction (NMSP), I summed all the statistics for all the players in the team's squad.

Experiment

Data Input

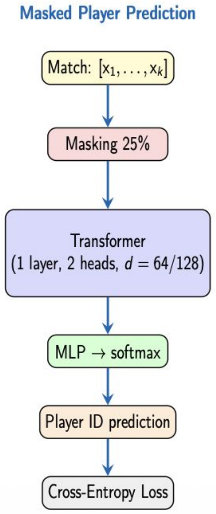
For each player, a vector $x \in \mathbb{R}^d$ is formed:



$$x = \underbrace{PE}_{\text{player ID}} + \underbrace{SPE}_{\text{position ID}} + \underbrace{TE}_{\text{team ID}} + \underbrace{TPE}_{\text{match stats}}$$

Augmentation: each match produces **10 samples** with different masks $\Rightarrow$ $\sim$18 000 training examples.

Architecture



Configurations: 1 layer, 2 heads, $d \in \{64, 128\}$

A) Learned position encoding

- Position ID is converted to one-hot.
- A learnable projection maps it to embedding space.

$$e_{\text{learned}}(p) = W_{\text{pos}} \cdot \mathrm{onehot}(p)$$

Experiment

3.1 Pre-Training with Masked Players Prediction (MPP)

Transformer models learn robust player representations through pretext tasks, which are crucial for achieving strong performance on downstream tasks. Inspired by the masked token prediction used in language modeling, I initially trained RisingBALLER using Masked Players Prediction (MPP). The goal of MPP is to enable the model to learn some prior relationships between players. The MPP task is straightforward: for each match, 25% of the input players are randomly masked, and the model is trained to predict these masked players.

I trained two models with one transformer layer each, using embedding dimensions of D=64 and D=128 respectively, across the 1,792 processed matches in the dataset. To enhance the model's generalization capabilities given the relatively small dataset size, I augmented the data by creating 10 different MPP inputs per match, each with different players masked. This resulted in approximately 18,000 input samples. The maximum sequence length per match was set to 80, and padding tokens were added if needed. I used 80% of the input samples for training and 20% for validation. The total number of tokens used for training was approximately 1.14 million, with a vocabulary size of 2,602, including the 2,600 unique players, the masking token, and the padding token.

The two models were trained with a batch size of 256, a learning rate of 0.0001, and a linear decay scheduler without warmup steps, using the AdamW [11] optimizer. The 64-dimensional model was trained for 280,000 steps (5,000 epochs), while the 128-dimensional model was trained for 56,000 steps (1,000 epochs). Following common practice in the literature, I used cross-entropy loss for training and evaluated top-1 and top-3 accuracy during training. The results are presented in Table 1.

Experiment

Architecture	**Split**	**Steps**	**Cross Entropy**	**Accuracy**	**Accuracy top3**
1l64d	train	280K	0.3873	x	x
1l64d	validation	280k	**0.8434**	**0.7893**	**0.9537**
1l128d	train	56K	0.3236	x	x
1l128d	validation	56K	0.8804	0.7764	0.9507

Experiment

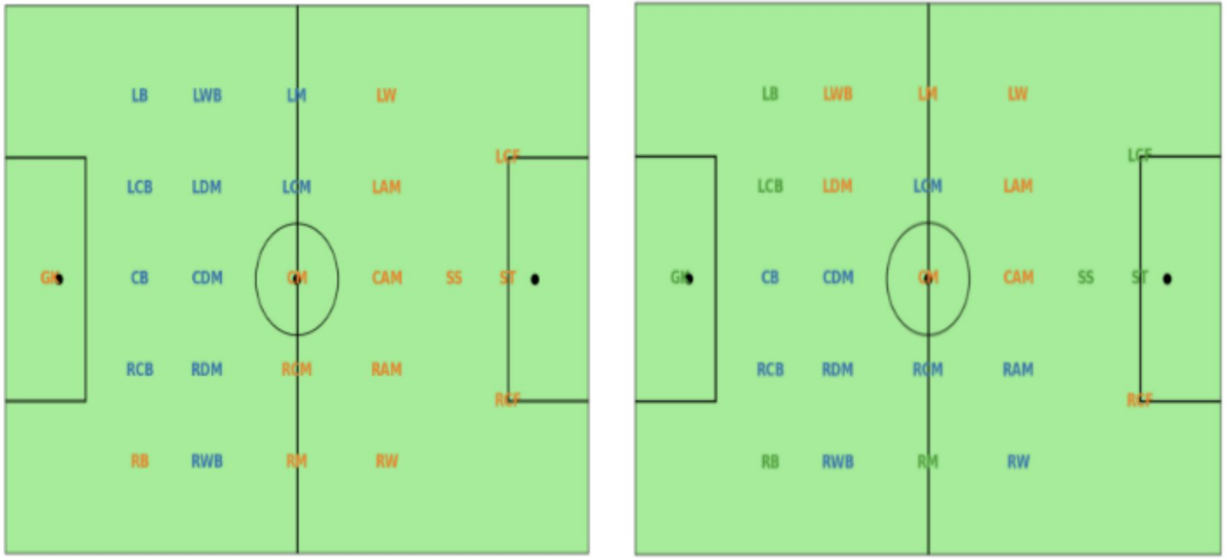


Figure 2: Left: Positional embeddings clustered into 2 groups. Right: Positional embeddings clustered into 3 groups.

Estimation

Table 4: The MPP pre-training results without teams affiliation embeddings; The clear drop in the accuracies for the two models showcases how harder it's for RisingBALLER to predict the masked players, by being forced to learn more difficult patterns for players embeddings with a small matches dataset.

Architecture	Split	Steps	Cross Entropy	Accuracy	Accuracy top3
1l64d	train	112K	1.2663	x	x
1l64d	validation	112k	**1.8417**	**0.4550**	**0.8006**
1l128d	train	112K	0.5402	x	x
1l128d	validation	112K	1.6708	0.5843	0.8830

Estimation

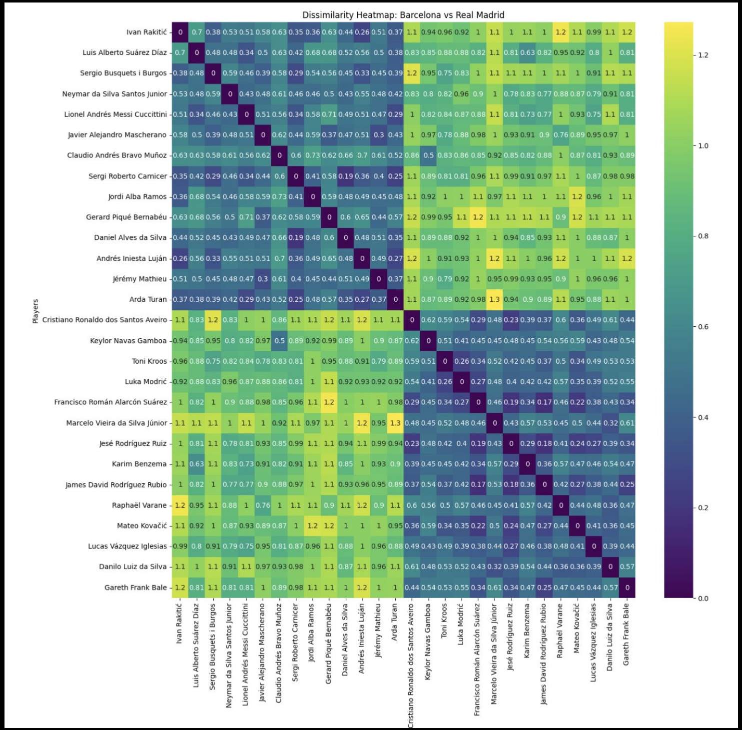


Table 8

Player of interest	Top 10 most similar players w TE	Top 10 most similar players w/o TE
Cristiano Ronaldo (RM)	Jesé Rodríguez Ruiz (RM) Francisco Román Isco (RM) Mateo Kovačić (RM) James Rodríguez (RM) Karim Benzema (RM) Borja Mayoral Moya (RM) Marcos Llorente Moreno (RM) Gareth Frank Bale (RM) Denis Cheryshev (Valencia) Nacho Fernández (RM)	Karim Benzema (Real Madrid) Neymar da Silva (Barcelona) Jesé Rodríguez (Real Madrid) Danilo da Silva (Real Madrid) Toni Kroos (Real Madrid) Imanol Arruti (Real Sociedad) Luis Suárez (Barcelona) Dani Ceballos (Real Betis) Gareth Bale (Real Madrid) Nacho Fernández (Real Madrid)

EventGPT (march 2026)

EventGPT: Capturing Player Impact from Team Action Sequences Using GPT-Based Framework

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Dataset

Table 1. Summary of the dataset used in experiments.

Property	Value
Seasons	2020/21 - 2024/25
Total Number of Matches	1,900
Total Number of Episodes	173,951
Average Number of Events per Episode	22.48
Total Number of Players	1,221

Experiment

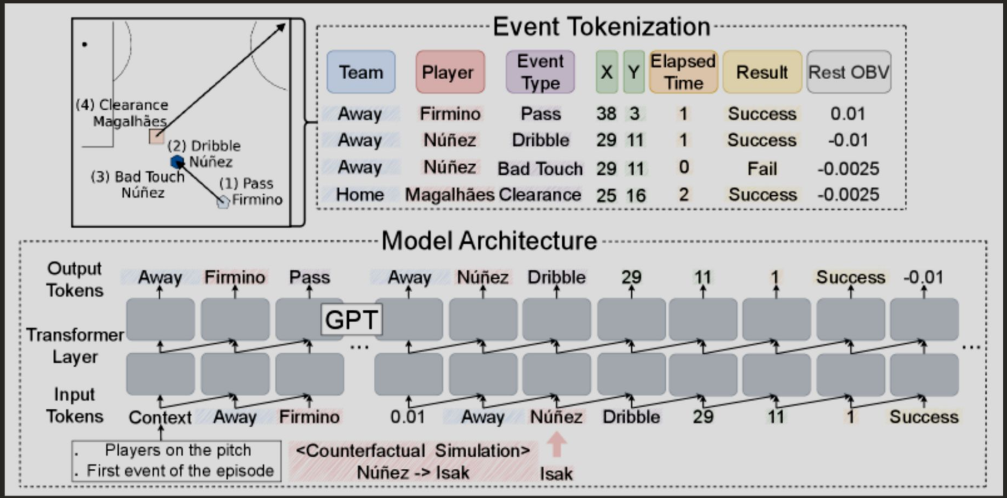


Figure 1. Overview of the EventGPT framework. Our nanoGPT-based Transformer model autoregressively predicts event tokens, enabling counterfactual 'what-if' simulations. For instance, replacing Darwin Núñez with Alexander Isak could alter actions (e.g., pass/shot) or modify the same action with a different location, outcome, or on-ball value (OBV). A video example of this sequence can be seen at the 23-second mark in the *linked video clip*.

Experiment

$$c = (pID_{1:22}, minute, h_{g'}, a_{g'}, h_{r'}, a_{r'}, h_{y'}, a_{y'}),$$

where $pID_{1:22}$ denotes the on-pitch players' ID, $minute$ is the current match minute, $h_{g'}, a_{g'}$ are cumulative goals, and h_r, a_r, h_y, a_y denote cumulative red/yellow cards for the home and away teams. This ensures that the model conditions event prediction on both who is on the pitch and the current match situation.

Experiment

- В отличие от xT, учитывает не только шансы забить, но и вероятность пропустить.

- Каждое действие оценивается по ΔxG (пропустить + забить)

- Группировка ΔxG по всем on-ball действиям по игрокам (set ~20)

- Нормирование в [0, 1] по позициям, получение “рейтинга” OBV игрока за матч + можно декомпозировать по действиям

- Усредненный OBV по матчам - “рейтинг” игрока за сезон

Experiment

SPADL (атрибуты действий)

АТРИБУТ	ОПИСАНИЕ
StartTime	Момент времени когда действие началось
EndTime	Момент времени когда действие закончилось
StartLoc	Координаты поля в начальный момент
EndLoc	Координаты поля в конечный момент
Player	Имя игрока выполняющего действие
Team	Название команды
ActionType	Тип действия (согласно градации на 21 действие)
BodyPart	Часть тела
Result	Результат действия (успешное или нет)

SPADL (типы действий)

ТИП ДЕЙСТВИЯ	ОПИСАНИЕ	УСПЕШНОСТЬ ДЕЙСТВИЯ	ДОПОЛНИТЕЛЬНЫЙ ТИП ИСХОДА
Pass	Пас на открытого игрока	Достигает партнера	Офсайд
Cross	Навес в штрафную	Достигает партнера	Офсайд
Throw-in	Вбрасывание (из аута)	Достигает партнера	
Crossed corner	Навес с углового в штрафную	Достигает партнера	Офсайд
Short corner	Розыгрыш углового	Достигает партнера	Офсайд
Crossed free-kick	Навес со штрафного в штрафную	Достигает партнера	Офсайд
Short free-kick	Розыгрыш штрафного	Достигает партнера	Офсайд
Take on	Попытка обыгрыша соперника	Сохраняется владение	
Foul	Фол	Всегда успешное	Желтая или красная
Tackle	Подкат	Восстановление владения	Желтая или красная
Interception	Перехват мяча	Всегда успешное	
Shot	Удар (не пенальти и не штрафной)	Гол	Автогол
Penalty shot	Пенальти	Гол	Автогол
Free-kick shot	Удар со штрафного	Гол	Автогол
Keeper save	Голкипер делает сейв после удара по воротам	Всегда успешное	
Keeper claim	Голкипер ловит мяч намертво после навеса (игра на выходах)	Не выронил мяч	
Keeper punch	Голкипер выбивает мяч после навеса (игра на выходах)	Всегда успешное	
Keeper pick-up	Голкипер забирает мяч в руки	Всегда успешное	
Clearance	Вынос мяча (безадресный)	Владение сохраняется	
Bad touch	Неудачный прием мяча, после которого мяч был потерян	Всегда неуспешное	
Dribble	Игрок проходит 3 и более метров с мячом (ведение мяча на дриблинге)	Всегда успешное	

Estimation

Table 2. Next-event prediction performance across baselines. Bold indicates top results, and underline indicates second-best results.

Model	h↑	e↑	x↓	y↓	t↓	a↑	rOBV↓
LEM	85.20%	74.07%	9.01	8.11	<u>1.37</u>	<u>90.51%</u>	0.014
LEM Transformer	**96.05%**	<u>80.42%</u>	<u>7.15</u>	<u>7.08</u>	1.40	86.92%	**0.008**
EventGPT	<u>94.12%</u>	**82.91%**	**4.30**	**4.31**	**1.11**	**92.87%**	<u>0.009</u>

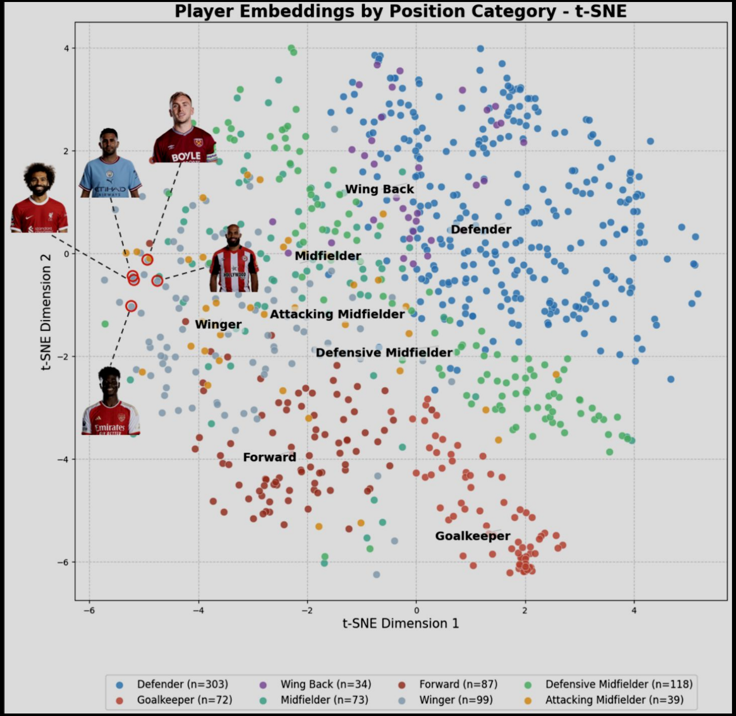
Notably, our model improves event type prediction accuracy compared to both baselines (82.91% vs. 80.42% for the LEM Transformer and 74.07% for LEM), indicating that conditioning on player identity and episode context leads to more accurate modeling of action selection. In addition, our model substantially reduces spatial prediction error, demonstrating improved understanding of how play unfolds on the field. While the LEM Transformer obtains the lowest rOBV error (0.008), EventGPT achieves a competitive second-best result (0.009).

We evaluate model performance across discrete and continuous event attributes using classification accuracy and mean absolute error (MAE), respectively.

- **Home/Away Team Indicator** (h_t): Accuracy
- **Event Type** (e_t): Accuracy
- **Spatial Coordinates** (x_t, y_t): MAE (measured in meters after rescaling)
- **Elapsed Time** (δ_t): MAE
- **Action Success Indicator** (o_t): Accuracy
- **Residual On-Ball Value** (rOBV_t): MAE

Accuracy is used for categorical prediction tasks, while continuous outputs are evaluated using MAE computed on the original (non-discretized) scale.

Application



Application

Player	Dribble (%)	Pass (%)	Shot (%)	Success Rate (%)	Pred. rOBV ↑
Alexander Isak (OBV: 4.94)	**64.34**	15.65	9.19	**81.83**	**3.76**
Rasmus Højlund (OBV: 1.80)	61.65	17.55	9.47	79.51	2.23
Darwin Núñez (OBV: 2.49)	61.82	18.01	9.64	80.57	2.12
Jean-Philippe Mateta (OBV: 0.65)	57.43	21.05	**10.54**	78.08	1.94
Erling Haaland (Sim) (OBV: 4.05)	57.30	**21.13**	10.39	77.93	2.71
Erling Haaland (GT)	59.35	18.11	12.33	78.42	2.59

Frontier AI

Всем спасибо за внимание!

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@MATCHCODE

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